

CS-AML MVP Engineering Breakdown v0.1

Engineering execution baseline for MVP 0.1

From SRS requirements to epics, stories, tasks, acceptance tests, dependencies, and release evidence.

Status: Baseline for engineering planning and backlog creation

1. Purpose and Engineering Objective

This document translates the approved CS-AML Product Requirements Document and Software Requirements Specification into an executable MVP engineering plan. It does not redefine product requirements. It establishes engineering work packages, sequencing, acceptance evidence, and a common definition of completion.

MVP release objective: a two-analyst team SHALL complete one sensitive investigation from case opening through independently reviewed and approved intelligence product, with provenance, authorization, and auditability intact.

1.1 Traceability chain

Framework → Product Feature → PRD → SRS → Epic → Story → Engineering Task → Acceptance Test → Release Evidence

1.2 Engineering principles

- Vertical slices over component completion.
- Canonical data first; search and graph remain rebuildable projections.
- Secure-by-default authorization is implemented in services/API, not only presentation logic.
- Analytical uncertainty is preserved in schema, UX, export, and tests.
- Every critical capability is accepted through observable behavior, not merely code presence.
- MVP integration scope is deliberately narrow; advanced OCR, sanctions, blockchain analytics, AI assistant and federation remain post-MVP unless required to unblock the release scenario.

2. MVP Scope Baseline

The P0 baseline is the mandatory release scope defined by the PRD/SRS. The engineering backlog groups those requirements into ten epics and eight delivery increments. P1/P2 capabilities are excluded unless explicitly promoted through change control.

Epic	Name	Release outcome	Primary traceability
E0	Platform Foundation & Delivery	Establish the deployable application skeleton, environments, CI/CD, canonical database, evidence storage, audit substrate, and engineering conventions.	SRS-OPS; SRS-DR; SRS-AUD; F-OPS-001; F-AUD-001
E1	Identity, Access & Protected Sources	Implement OIDC/MFA integration, role/object access controls, case membership, classification, and protected-source compartment.	SRS-SEC; F-SEC-001..003
E2	Case & Investigation Workflow	Implement case register, investigation charter, lifecycle gates, tasks, assignments, and case activity history.	SRS-FR-CASE; F-CASE-001..005

Epic	Name	Release outcome	Primary traceability
E3	Source, Evidence & Document Intake	Implement source provenance, original evidence preservation, hashing, extracts, derivative lineage, ratings, and document ingestion.	SRS-FR-EVD/DOC; F-EVD-001..006; F-DOC-001
E4	Entity Resolution & Relationship Model	Implement entity registry, aliases/identifiers, candidate matching, merge/unmerge, relationship, ownership/control, and asset registry.	SRS-FR-ENT/REL/AST; F-ENT-001..005; F-REL-001..003; F-AST-001
E5	Timeline & Follow-the-Value	Implement events, visual timeline, ValueFlow records, multi-leg chains, uncertainty classes, range/unknown values, and flow visualization.	SRS-FR-TIM/VAL; F-TIM-001..002; F-VAL-001..004
E6	Typology, Hypothesis & Assessment	Implement typology catalogue, indicators, typology worksheet, competing hypotheses, gaps, confidence, and assessments.	SRS-FR-TYP/HYP/ASM; F-TYP-001..003; F-HYP-001..003; F-ASM-001..003
E7	Search & Investigation Graph	Implement permission-aware full-text/faceted search and evidence-backed graph visualization with canonical links.	SRS-FR-SCH/GRF; F-SCH-001..002; F-GRF-001
E8	Intelligence Product, Review & Dissemination	Implement report templates, evidence index, peer review, versioning/corrections, approvals, export/referral, and dissemination log.	SRS-FR-PRD/REV/DIS; F-PRD-001..003; F-REV-001; F-DIS-001..004
E9	Administration, Retention & Operational Readiness	Implement vocabularies, retention policies, backup/restore, configuration, minimum monitoring, security hardening, and release verification.	SRS-FR-ADM/SEC/AUD/OPS; F-ADM-001; F-ADM-003; F-OPS-001

3. Delivery Strategy and Increments

The recommended sequence creates demonstrable end-to-end capabilities early. Increments are dependency-aware, but teams MAY overlap work where contracts are stable and integration tests remain authoritative.

Increment	Primary scope	Exit outcome
I0 — Engineering Foundation	E0 + core E1	Runnable secure skeleton; migrations, canonical DB, object storage, audit, OIDC baseline.
I1 — Governed Case Workspace	E1 + E2	Authenticated team can create

Increment	Primary scope	Exit outcome
		compartmentalized case, charter, gates, tasks and activity.
I2 — Evidence-to-Entity Chain	E3 + E4 partial	Team can register source, preserve evidence, cite extracts and build reviewable entities.
I3 — Investigation Model	E4 remainder + E5	Relationships, assets, timeline and value-flow analysis become usable end-to-end.
I4 — Analytical Reasoning	E6	Typology, hypotheses, gaps, assessment and confidence complete analytical chain.
I5 — Discovery & Graph	E7	Search and graph improve navigation without becoming canonical truth.
I6 — Reviewed Intelligence Product	E8	Assessment becomes reviewed/versioned intelligence product with controlled export.
I7 — Operational Release	E9 + regression	Retention, restore, hardening and full MVP release scenario verified.

4. Dependency Model

Core dependency direction: E0 precedes all epics; E1 is required before sensitive data becomes multi-user; E2/E3 establish governed investigation context and evidence; E4/E5 create the financial investigation model; E6 creates analytical reasoning; E7 accelerates discovery; E8 creates controlled intelligence outputs; E9 closes operational release risk.

Critical path: E0 → E1 → E2 → E3 → E4 → E5 → E6 → E8 → E9. E7 can begin after E4 canonical relationships are stable.

5. Detailed Engineering Backlog

E0 — Platform Foundation & Delivery

Establish the deployable application skeleton, environments, CI/CD, canonical database, evidence storage, audit substrate, and engineering conventions.

Traceability: SRS-OPS; SRS-DR; SRS-AUD; F-OPS-001; F-AUD-001

ST-E0-01 — Repository and service skeleton

As an engineer, I need a reproducible project structure so all later capabilities share conventions.

Engineering tasks

- Create backend, frontend, worker and infrastructure directories
- Configure environment-based settings and secrets
- Define coding/lint/test conventions
- Add health endpoint and build metadata

Acceptance tests / exit criteria

- Fresh checkout can run locally from documented commands
- CI executes lint + unit tests
- No production secret is committed

ST-E0-02 — Canonical PostgreSQL baseline

As a data steward, I need versioned canonical storage so analytical objects have a durable source of truth.

Engineering tasks

- Create database migration framework
- Implement common object envelope: UUID, timestamps, version, classification, created_by
- Add soft/superseded state conventions
- Create migration rollback/test strategy

Acceptance tests / exit criteria

- Migrations apply to empty database
- Schema version is queryable
- Canonical IDs remain stable across updates

ST-E0-03 — Evidence object storage baseline

As an evidence custodian, I need originals stored separately from metadata.

Engineering tasks

- Configure S3-compatible or filesystem evidence store
- Use opaque object keys, not user filenames
- Store checksum and content metadata in DB
- Block in-place overwrite of originals

Acceptance tests / exit criteria

- Uploading same filename twice creates distinct evidence versions
- Original object cannot be modified by analyst role

ST-E0-04 — Audit event substrate

As an auditor, I need material changes captured consistently.

Engineering tasks

- Define AuditEvent schema
- Create application audit helper/middleware
- Capture actor, action, object, before/after reference, timestamp, request correlation
- Protect audit records from ordinary modification

Acceptance tests / exit criteria

- Create/update/delete/merge/export actions produce events
- Analyst cannot alter audit history

E1 — Identity, Access & Protected Sources

Implement OIDC/MFA integration, role/object access controls, case membership, classification, and protected-source compartment.

Traceability: SRS-SEC; F-SEC-001..003

ST-E1-01 — OIDC login and session policy

As a user, I need organisational SSO so access can be centrally revoked.

Engineering tasks

- Implement OIDC authorization code flow
- Map IdP subject to local user
- Enforce session timeout and logout
- Support MFA claim/policy checks

Acceptance tests / exit criteria

- Disabled IdP user cannot establish new session
- Session shows authenticated identity and roles

ST-E1-02 — Role and case membership authorization

As a case owner, I need need-to-know access so sensitive cases stay compartmentalised.

Engineering tasks

- Define roles: investigator, case_owner, reviewer, data_steward, evidence_custodian, admin, auditor
- Implement case membership ACL
- Apply authorization in API layer, not UI only
- Add classification checks

Acceptance tests / exit criteria

- Direct API request cannot bypass case membership
- Unauthorized object IDs return non-disclosing denial

ST-E1-03 — Protected-source compartment

As a source handler, I need source identity separated from routine evidence.

Engineering tasks

- Create ProtectedSource model/store

- Separate identity from source-derived evidence reference
- Restrict handler role
- Redact identity from normal search/index/audit payloads

Acceptance tests / exit criteria

- Routine analyst can use source-derived evidence without source identity
- Protected identity never appears in normal export

E2 — Case & Investigation Workflow

Implement case register, investigation charter, lifecycle gates, tasks, assignments, and case activity history.

Traceability: SRS-FR-CASE; F-CASE-001..005

ST-E2-01 — Case register and creation

As an investigator, I need a stable case workspace.

Engineering tasks

- Create Case model/API/UI
- Required fields: title, purpose, owner, classification, status
- Generate stable case identifier
- Case list filters by status/owner/classification

Acceptance tests / exit criteria

- Case cannot activate without owner, purpose, question
- Case history records creation

ST-E2-02 — Investigation Charter

As a case owner, I need scope and boundaries recorded before substantive work.

Engineering tasks

- Implement investigation question, scope included/excluded, jurisdictions, period, risks, authorised/prohibited collection
- Version charter
- Require reason for material scope change

Acceptance tests / exit criteria

- Old charter version remains retrievable
- Scope change produces audit event

ST-E2-03 — Lifecycle gates G0-G6

As governance reviewer, I need high-risk actions controlled by gates.

Engineering tasks

- Model gates and gate requirements
- Gate submission/approval/rejection
- Prevent self-approval when policy requires independence
- Record conditions and unresolved gaps

Acceptance tests / exit criteria

- Controlled transition is blocked without approval
- Approval predates downstream action

ST-E2-04 — Tasks and assignments

As an investigator, I need actionable work tracked.

Engineering tasks

- Task model with owner/status/due date/type
- Case task board/list
- Completion history
- Link tasks to sources/entities/hypotheses where relevant

Acceptance tests / exit criteria

- Task ownership and completion are visible
- Closed task retains historical state

ST-E2-05 — Case activity timeline

As a reviewer, I need to reconstruct material case activity.

Engineering tasks

- Project relevant AuditEvents into case activity
- Filter by actor/action/object
- Link activity to canonical object

Acceptance tests / exit criteria

- Material case events appear chronologically
- Activity item opens the referenced object when authorized

E3 — Source, Evidence & Document Intake

Implement source provenance, original evidence preservation, hashing, extracts, derivative lineage, ratings, and document ingestion.

Traceability: SRS-FR-EVD/DOC; F-EVD-001..006; F-DOC-001

ST-E3-01 — Source register

As an investigator, I need material sources registered with provenance.

Engineering tasks

- Source model/API/UI
- Capture origin, publisher, URL/location, publication/access dates, access method, legal note
- Archive reference and reliability fields

Acceptance tests / exit criteria

- Assessment-linked source must contain provenance minimum
- Mutable web source can reference preserved copy

ST-E3-02 — Original evidence ingestion

As a custodian, I need evidence originals preserved.

Engineering tasks

- Upload API with streaming size limits
- Virus/malware scanning hook
- Write immutable original
- Capture MIME/size/filename/acquired_at/collector

Acceptance tests / exit criteria

- Failed ingest never creates misleading completed record
- Original remains unchanged after analyst actions

ST-E3-03 — Hash/integrity verification

As a reviewer, I need to verify evidence integrity.

Engineering tasks

- Compute SHA-256 at ingest
- Store algorithm/hash
- Add reverify action
- Flag mismatch

Acceptance tests / exit criteria

- Recomputed hash matches stored original in normal case
- Mismatch creates security/audit event

ST-E3-04 — Evidence extracts and citations

As an analyst, I need precise excerpts linked to originals.

Engineering tasks

- EvidenceExtract object
- Support page/paragraph/region/location descriptor
- Preview parent context
- Link extract to claims/facts

Acceptance tests / exit criteria

- Extract cannot exist without parent evidence
- Citation opens exact parent location where supported

ST-E3-05 — Derivative lineage

As an analyst, I need OCR/translation/processed copies distinguished from originals.

Engineering tasks

- Derivative model and parent link
- Transformation type/tool/version/time/creator
- Prevent derivative from replacing original
- Lineage UI

Acceptance tests / exit criteria

- Every derivative traces to original
- Original and derivative are visually distinct

ST-E3-06 — Reliability and credibility ratings

As an analyst, I need source reliability separate from information credibility.

Engineering tasks

- Implement A-F source scale
- Implement 1-6 information credibility scale
- Require rationale for material ratings
- Allow reassessment history

Acceptance tests / exit criteria

- UI does not merge the two ratings
- Rating change is versioned/audited

E4 — Entity Resolution & Relationship Model

Implement entity registry, aliases/identifiers, candidate matching, merge/unmerge, relationship, ownership/control, and asset registry.

Traceability: SRS-FR-ENT/REL/AST; F-ENT-001..005; F-REL-001..003; F-AST-001

ST-E4-01 — Entity registry

As an analyst, I need reusable entities outside case silos.

Engineering tasks

- Entity model by type
- Canonical name, aliases, identifiers, jurisdictions, status
- Case-to-entity association without duplicating entity

Acceptance tests / exit criteria

- Same canonical entity can appear in multiple authorized cases
- Entity record preserves provenance-bearing assertions

ST-E4-02 — Aliases and identifiers

As an analyst, I need conflicting identifiers represented without destructive overwrite.

Engineering tasks

- Identifier records with type/value/source/status
- Alias records with source/time
- Uniqueness rules only where justified

Acceptance tests / exit criteria

- Conflicting values coexist with provenance
- No silent overwrite

ST-E4-03 — Candidate entity matching

As an analyst, I need a reviewable duplicate-candidate screen.

Engineering tasks

- Candidate matching on normalized name + identifiers
- Display matching and conflicting attributes
- No auto-merge in MVP

Acceptance tests / exit criteria

- Candidate screen explains why pair was suggested
- Common name alone does not force merge

ST-E4-04 — Merge and unmerge

As a data steward, I need reversible entity resolution.

Engineering tasks

- MergeDecision record

- Preserve old IDs as aliases/superseded records
- Repoint relationships with history
- Implement unmerge recovery

Acceptance tests / exit criteria

- Unmerge restores prior object topology
- Merge rationale/evidence is mandatory

ST-E4-05 — Relationships, ownership and control

As an analyst, I need first-class connections with evidence.

Engineering tasks

- Relationship model with type/endpoints/evidence/confidence/time
- OwnershipInterest with percentage nullable
- ControlAssertion distinguishes legal owner/beneficial owner/controller/user

Acceptance tests / exit criteria

- Graph edge always traces to relationship record
- Unknown percentage remains null, not zero

ST-E4-06 — Asset registry

As an analyst, I need assets linked to ownership/control assertions.

Engineering tasks

- Asset model/types/identifiers/location
- Attribution relationship to person/org
- Evidence and status

Acceptance tests / exit criteria

- Asset attribution distinguishes ownership/control/use
- Material attribution requires evidence link

E5 — Timeline & Follow-the-Value

Implement events, visual timeline, ValueFlow records, multi-leg chains, uncertainty classes, range/unknown values, and flow visualization.

Traceability: SRS-FR-TIM/VAL; F-TIM-001..002; F-VAL-001..004

ST-E5-01 — Events and temporal precision

As an analyst, I need chronology with uncertainty.

Engineering tasks

- Event object with date precision
- Link entities/assets/evidence
- Support exact date, month, year, range, unknown

Acceptance tests / exit criteria

- Approximate date is not rendered as exact
- Event links to supporting evidence

ST-E5-02 — Visual timeline

As an analyst, I need case chronology to reveal patterns.

Engineering tasks

- Timeline query/API
- Filters by entity/type/date
- Interactive UI and export representation

Acceptance tests / exit criteria

- Timeline respects object permissions
- Each item opens canonical event/evidence

ST-E5-03 — ValueFlow canonical model

As an analyst, I need transfer of economic value represented without pretending all flows are transactions.

Engineering tasks

- ValueFlow fields origin/destination/mechanism/value/currency/date/evidence
- Mandatory flow_class enum
- Confidence/status

Acceptance tests / exit criteria

- Cannot save flow without class
- Unknown value allowed without zero substitution

ST-E5-04 — Multi-leg flow builder

As an analyst, I need reconstructable chains.

Engineering tasks

- ValueFlowLeg model/order
- Builder UI
- Per-leg evidence/confidence/class
- Chain validation

Acceptance tests / exit criteria

- Each leg can carry different evidence/class
- Removing a leg does not destroy evidence objects

ST-E5-05 — Flow visualisation

As a reviewer, I need direct vs reconstructed flows visually unmistakable.

Engineering tasks

- Distinct labels/line semantics for four classes
- Legend always visible
- Export keeps class labels

Acceptance tests / exit criteria

- Screenshot/export can distinguish all classes without color alone
- No derived flow shown as direct

E6 — Typology, Hypothesis & Assessment

Implement typology catalogue, indicators, typology worksheet, competing hypotheses, gaps, confidence, and assessments.

Traceability: SRS-FR-TYP/HYP/ASM; F-TYP-001..003; F-HYP-001..003; F-ASM-001..003

ST-E6-01 — Typology catalogue browser

As an analyst, I need versioned CS-AML typologies in the workflow.

Engineering tasks

- Load catalogue data
- Browse/search typologies
- Display mechanism, observables, false positives, version

Acceptance tests / exit criteria

- Assessment records catalogue version used
- Typology content cannot silently mutate historical case

ST-E6-02 — Indicator capture

As an analyst, I need evidence-linked indicators and counter-indicators.

Engineering tasks

- Indicator model with direction/type/significance
- Links to entities/events/flows/evidence
- Counter-indicator support

Acceptance tests / exit criteria

- Material indicator has evidence/proposition
- Indicator itself is not labelled proof

ST-E6-03 — Typology match worksheet

As an analyst, I need structured comparison rather than automatic accusation.

Engineering tasks

- TypologyMatch object
- Observed indicators/counter-indicators
- Consistency level
- Alternative explanations

Acceptance tests / exit criteria

- Single weak indicator cannot produce strong level automatically
- Analyst rationale required

ST-E6-04 — Competing hypotheses

As an analyst, I need multiple explanations tested.

Engineering tasks

- Hypothesis object/status
- Supporting/contradicting evidence links
- Alternative hypothesis set
- Assumptions

Acceptance tests / exit criteria

- At least two plausible hypotheses supported in pilot
- Rejected hypothesis remains in history

ST-E6-05 — Intelligence gaps

As an analyst, I need unknowns explicit.

Engineering tasks

- Gap object/question/impact/priority/status
- Link to hypothesis/assessment
- Closure rationale

Acceptance tests / exit criteria

- Assessment can list unresolved gaps
- Gap cannot be silently deleted

ST-E6-06 — Assessment and confidence

As an analyst, I need judgement with basis and uncertainty.

Engineering tasks

- Assessment versioning
- Judgement, confidence, rationale, assumptions, gaps, alternatives
- Evidence traversal

Acceptance tests / exit criteria

- Reviewer can trace assessment backward to evidence
- Confidence basis is mandatory for material assessment

E7 — Search & Investigation Graph

Implement permission-aware full-text/faceted search and evidence-backed graph visualization with canonical links.

Traceability: SRS-FR-SCH/GRF; F-SCH-001..002; F-GRF-001

ST-E7-01 — Permission-aware global search

As an analyst, I need to find authorized entities, cases and evidence.

Engineering tasks

- Search service/index or DB FTS
- Permission filtering before result display
- Filters by object type/case/date

Acceptance tests / exit criteria

- Unauthorized object never appears in count/snippet
- Search result links canonical object

ST-E7-02 — Faceted case search

As an analyst, I need focused search within a case.

Engineering tasks

- Facets for source/entity/event/evidence/relationship
- Case scope enforcement
- Saved query not required in MVP

Acceptance tests / exit criteria

- Facets respect ACL and classification
- Query response time meets MVP NFR target

ST-E7-03 — Investigation graph

As an analyst, I need visual entity/relationship/asset/value-flow exploration.

Engineering tasks

- Graph projection from canonical data
- Node/edge filtering
- Evidence/provenance side panel
- Save view state optionally

Acceptance tests / exit criteria

- Every material edge links to canonical relationship/evidence
- Projection can be rebuilt without data loss

E8 — Intelligence Product, Review & Dissemination

Implement report templates, evidence index, peer review, versioning/corrections, approvals, export/referral, and dissemination log.

Traceability: SRS-FR-PRD/REV/DIS; F-PRD-001..003; F-REV-001; F-DIS-001..004

ST-E8-01 — Intelligence product templates

As an analyst, I need consistent report outputs.

Engineering tasks

- Product model
- Templates: Financial Intelligence Note, Referral Package, Case Report
- Auto-populate metadata/classification/evidence index references

Acceptance tests / exit criteria

- Draft is versioned
- Product shows author/reviewer/status

ST-E8-02 — Evidence index generation

As a reviewer, I need claims traceable from report to evidence.

Engineering tasks

- Generate index from linked facts/assessments
- Include source/evidence IDs and citation locations
- Permission-aware inclusion

Acceptance tests / exit criteria

- Key fact in pilot has traversable evidence reference
- Restricted evidence is handled according to export policy

ST-E8-03 — Peer review workflow

As a reviewer, I need independent approval and change requests.

Engineering tasks

- Review states/commenting/request-changes/approve
- Prevent author self-approval where required
- Freeze reviewed version

Acceptance tests / exit criteria

- Approved version identifies independent reviewer
- Post-approval change creates new version

ST-E8-04 — Corrections and supersession

As a case owner, I need prior products retained when corrected.

Engineering tasks

- Supersedes relationship
- Status current/superseded/retracted
- Correction note

Acceptance tests / exit criteria

- Old version remains auditable
- Recipient-facing export identifies current version

ST-E8-05 — Dissemination approval and export

As a case owner, I need controlled external release.

Engineering tasks

- Dissemination object recipient/purpose/classification/approval
- Export minimization selector
- Generate package/report

Acceptance tests / exit criteria

- External export blocked before approval
- Export manifest lists included objects/version

ST-E8-06 — Referral package and sharing log

As an investigator, I need structured referral plus immutable sharing record.

Engineering tasks

- Referral template sections

- Sharing log record recipient/time/version/restrictions
- Audit export event

Acceptance tests / exit criteria

- Referral separates fact/analysis/gaps
- Ordinary analyst cannot edit completed sharing log

E9 — Administration, Retention & Operational Readiness

Implement vocabularies, retention policies, backup/restore, configuration, minimum monitoring, security hardening, and release verification.

Traceability: SRS-FR-ADM/SEC/AUD/OPS; F-ADM-001; F-ADM-003; F-OPS-001

ST-E9-01 — Controlled vocabulary admin

As an admin, I need versioned vocabularies.

Engineering tasks

- Relationship/asset/status/classification vocabulary UI
- Version terms
- Disable without deleting historical semantics

Acceptance tests / exit criteria

- Historical record displays original term/version
- Vocabulary change audited

ST-E9-02 — Retention policy engine

As a privacy/admin role, I need retention/disposition states.

Engineering tasks

- RetentionPolicy model
- Apply by classification/object type
- Preview eligible records
- Hold/exception support

Acceptance tests / exit criteria

- Disposition requires authorization
- Legal hold blocks deletion

ST-E9-03 — Backup and restore

As an operator, I need recoverable canonical data and evidence.

Engineering tasks

- Automated DB backup
- Evidence store backup strategy
- Configuration backup
- Restore runbook and test

Acceptance tests / exit criteria

- Documented restore test succeeds
- Restored hashes/evidence references validate

ST-E9-04 — Operational hardening

As an operator, I need production-safe defaults.

Engineering tasks

- Security headers/TLS/reverse proxy
- Secrets management
- Rate limits/upload limits
- Error handling without sensitive leakage
- Minimal health/metrics

Acceptance tests / exit criteria

- No debug mode in production
- Sensitive evidence content absent from routine logs

ST-E9-05 — MVP end-to-end release test

As product owner, I need objective release evidence.

Engineering tasks

- Create representative pilot dataset
- Automate what can be automated
- Run two-analyst scenario
- Record failures and remediation

Acceptance tests / exit criteria

- All mandatory P0 SRS requirements pass or accepted exception exists
- Pilot completes case-to-approved-product with intact provenance/audit

6. Cross-Cutting Engineering Requirements

6.1 Authorization

- Every API read/write decision is evaluated server-side.

- Object-level classification and case membership are evaluated before data projection or export.
- Search/graph must not leak inaccessible object existence through counts, snippets, or topology.

6.2 Provenance and audit

- All canonical analytical objects expose stable IDs.
- Material mutations record actor/time/reason and relevant before/after references.
- Evidence extracts, derivatives, facts, indicators, hypotheses and assessments remain traversable backward to sources.

6.3 Data integrity

- Original evidence cannot be edited in place.
- Entity merge is reversible.
- ValueFlow class is mandatory and cannot be lost in export.
- Unknown numeric values remain unknown; null is not coerced to zero.

6.4 Testing

- Unit tests cover domain invariants.
- Integration tests cover database/storage/auth boundaries.
- Authorization tests use positive and negative cases.
- End-to-end test follows the Annex release scenario.
- Backup restore is a release test, not only an ops procedure.

7. Suggested Repository and Module Boundaries

The following structure is illustrative and SHOULD be adjusted to the selected stack without collapsing domain boundaries:

```
cs-aml/
  backend/
    apps/cases
    apps/evidence
    apps/entities
    apps/analysis
    apps/products
    apps/governance
    apps/audit
  frontend/
  worker/
  infra/
  tests/
    unit/
    integration/
    e2e/
```

security/
docs/

8. Engineering Definition of Ready

- Story has SRS/feature traceability.
- Data objects and authorization scope are known.
- Acceptance criteria are testable.
- Required upstream APIs/contracts are stable or mocked.
- Security/privacy impact is identified.
- Open design decision that could invalidate implementation is resolved or explicitly timeboxed.

9. Engineering Definition of Done

- Code reviewed and merged through protected branch workflow.
- Automated unit/integration tests pass.
- Negative authorization tests pass where applicable.
- Audit/provenance behavior is verified.
- Schema/API documentation updated.
- No critical/high unresolved security defect for the capability.
- Acceptance criteria demonstrated against representative data.
- Feature traceability status updated.

10. MVP Release Evidence Package

- SRS traceability matrix showing implementation and verification status.
- End-to-end two-analyst pilot record.
- Access-control and protected-source compartment test results.
- Evidence hash, original/derivative lineage and citation tests.
- Entity merge/unmerge recovery test.
- Value-flow class persistence across DB/API/UI/export.
- Hypothesis and assessment traceability test.
- Independent peer-review and dissemination-approval test.
- Backup/restore test record.
- Known limitations and formally accepted exceptions.

11. Post-MVP Deferred Work

The following capabilities remain outside the mandatory MVP unless a release dependency is discovered: automated OCR pipelines, entity NLP extraction, OpenSanctions screening, OpenAleph connector, FollowTheMoney import/export, external graph database, advanced path/network

analytics, GraphSense, Flowintel interoperability, AI summarisation/assistant, partner federation, multi-tenant hosting, advanced alerting and watch queries.

Annex A — Epic Exit Checklist

Epic	Scope	Exit rule
E0	Platform Foundation & Delivery	All stories accepted; traceability updated; domain integration test passes; no unresolved critical security/data-integrity defect.
E1	Identity, Access & Protected Sources	All stories accepted; traceability updated; domain integration test passes; no unresolved critical security/data-integrity defect.
E2	Case & Investigation Workflow	All stories accepted; traceability updated; domain integration test passes; no unresolved critical security/data-integrity defect.
E3	Source, Evidence & Document Intake	All stories accepted; traceability updated; domain integration test passes; no unresolved critical security/data-integrity defect.
E4	Entity Resolution & Relationship Model	All stories accepted; traceability updated; domain integration test passes; no unresolved critical security/data-integrity defect.
E5	Timeline & Follow-the-Value	All stories accepted; traceability updated; domain integration test passes; no unresolved critical security/data-integrity defect.
E6	Typology, Hypothesis & Assessment	All stories accepted; traceability updated; domain integration test passes; no unresolved critical security/data-integrity defect.
E7	Search & Investigation Graph	All stories accepted; traceability updated; domain integration test passes; no unresolved critical security/data-integrity defect.
E8	Intelligence Product, Review & Dissemination	All stories accepted; traceability updated; domain integration test passes; no unresolved critical security/data-integrity defect.
E9	Administration, Retention & Operational Readiness	All stories accepted; traceability updated; domain integration test passes; no unresolved critical security/data-integrity defect.

Annex B — MVP End-to-End Acceptance Scenario

1. Authenticate two analysts and one independent reviewer through configured identity provider.
2. Create classified case, owner, investigation question and charter; complete required gate.

3. Register a public source and ingest original PDF evidence; compute hash and create precise extract.
4. Create two entities and one organization; record aliases/identifiers; exercise candidate match and reviewed merge/unmerge on test records.
5. Create evidence-backed relationship, ownership/control assertion and an asset.
6. Add events and render chronological timeline.
7. Create DIRECT/DOCUMENTED and RECONSTRUCTED/HYPOTHETICAL value-flow examples and verify visual/export distinction.
8. Map indicators to one typology, record counter-indicator, and create at least two competing hypotheses.
9. Create intelligence gap and confidence-rated assessment linked backward to evidence.
10. Use search and graph to retrieve only objects authorized to the case team.
11. Generate intelligence product and evidence index; submit independent peer review; revise if required.
12. Approve dissemination and create controlled export/referral package; verify sharing log and audit event.
13. Attempt unauthorized access from non-member account and confirm non-disclosing denial.
14. Run backup and restore test; verify evidence hashes and canonical references after restore.

Annex C — Initial Engineering Decision Log

Decision	Topic	Guidance	Status
ED-01	Backend framework	Select implementation stack compatible with SRS and architecture; default reference profile may use Django/DRF.	Open until kickoff
ED-02	Frontend	Select SPA/server-rendered approach; must support graph/timeline interactions and accessible forms.	Open until kickoff
ED-03	Evidence storage	S3-compatible object store vs hardened filesystem.	Decide before E0-ST03
ED-04	Search	PostgreSQL FTS for MVP vs external OpenSearch.	Recommend PostgreSQL FTS for MVP unless scale requires otherwise
ED-05	Graph	Relational/derived graph in MVP vs Neo4j/Memgraph.	Recommend derived relational/API graph for MVP; external graph post-MVP

Decision	Topic	Guidance	Status
ED-06	IAM	OIDC provider and MFA policy.	Required before E1
ED-07	Deployment	Single-organisation deployment profile for MVP.	Recommended